Exploratory Data Analysis Summary Report

Geldium Credit Customer Dataset — Data Quality Audit, Missing Value Treatment & Delinquency Risk Indicator Analysis

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1. Introduction

This report presents the findings of an exploratory data analysis conducted on Geldium's credit customer dataset. The objective is to evaluate the quality and completeness of the data before it is used to build a predictive delinquency risk model. Our review focuses on identifying missing values, spotting inconsistencies, and surfacing early indicators of customer default risk.

The analysis was carried out using a combination of Python (Pandas), Microsoft Excel, and generative AI tools. The insights documented here are intended to guide the analytics team in refining their data pipeline and to help Geldium's leadership understand the current state of their customer data.

2. Dataset Overview

The dataset contains **393 customer records** across 19 columns. Each record captures a customer's demographic profile, financial standing, credit product details, and a six-month payment history. The target variable, `Delinquent_Account`, is a binary flag (0 = No, 1 = Yes) indicating whether a customer's account is currently delinquent.

Key Variables

Variable	Type	Description
Customer_ID	Categorical	Unique alphanumeric identifier for each customer
Age	Numerical	Customer age in years (range: 18–74)
Income	Numerical	Annual income in USD; *contains 28 missing values*
Credit_Score	Numerical	Credit score (range: 301–847); *contains 2 missing values*
Credit_Utilization	Numerical	Proportion of available credit in use (0.05–1.03)
Missed_Payments	Numerical	Number of missed payments in the past 12 months (0–6)
Delinquent_Account	Binary	Target variable: 0 = No, 1 = Yes
Loan_Balance	Numerical	Outstanding loan balance in USD; *contains 21 missing values*
Debt_to_Income_Ratio	Numerical	Total debt divided by annual income
Employment_Status	Categorical	Employment status (contains duplicate categories)
Account_Tenure	Numerical	Number of years the account has been active (0–19)
Credit_Card_Type	Categorical	Card tier: Standard, Gold, Platinum, Business, Student
Location	Categorical	City of residence (5 cities)
Month_1 to Month_6	Categorical	Monthly payment status: On-time, Late, or Missed

Anomalies & Inconsistencies Identified

1. **Credit utilization above 100%:** Records such as CUST0090 (102.6%), CUST0266 (102.5%), and CUST0293 (100.2%) show utilization rates exceeding the theoretical maximum. This can happen when interest or fees push the balance above the credit limit, but these values should be capped at 1.0 before modeling to avoid distortion.
2. **Employment status formatting:** The same employment category appears under multiple labels— `EMP` , `employed` , and `Employed` all refer to the same status. Similarly, `retired` appears in lowercase only. These need to be

standardized into a consistent set of categories before any encoding takes place.

3. **Credit card type overlap:** The `Credit_Card_Type` column includes “Student” and “Business” alongside traditional card tiers. This creates potential confusion with the employment status column and could introduce multi-collinearity if both are used as features.

📝 Initial Data Quality Summary

The Geldium dataset provides a reasonable foundation for analysis with 393 records and a diverse mix of financial and behavioral features. However, it requires targeted cleaning before it can support reliable predictions. Missing values affect three critical columns—Income, Credit Score, and Loan Balance—and the employment status field contains inconsistent text labels that would cause category duplication in any model. Additionally, a small number of credit utilization values exceed 100%, representing edge cases that need to be handled. With the cleaning steps outlined in this report, the dataset should be ready for feature engineering and model training.

3. Missing Data Analysis

Three variables contain missing values. Left untreated, these gaps would either cause model training to fail entirely or introduce bias into the predictions. The table below summarizes each issue, the chosen treatment approach, and the reasoning behind it.

Variable	Missing Count	Treatment Method	Justification
Income	28 (7.1%)	Group-based median imputation (by employment status)	Income levels differ substantially across employment groups. A global median would overestimate income for unemployed customers and underestimate it for employed ones. Grouping by employment status preserves the natural spread of the data.
Credit_Score	2 (0.5%)	Global median imputation (median = 579)	With only two missing records, the impact of any imputation method is negligible. Median imputation is the simplest and most defensible choice for such a small gap.
Loan_Balance	21 (5.3%)	Global median imputation (median = $44,454)	Loan balances are right-skewed, with a few very large values. The median is more resistant to these outliers than the mean, making it a better central estimate for imputation.

Excel Formulas for Imputation

For analysts working directly in Excel, the following formulas can be applied in a new column adjacent to the original data:

Field	Excel Formula
Income	`=IF(ISBLANK(C2), VLOOKUP(J2, MedianIncomeTable, 2, FALSE), C2)` *Requires a helper table with median income by employment status.*
Credit Score	`=IF(ISBLANK(D2), MEDIAN(D$2:D$394), D2)`
Loan Balance	`=IF(ISBLANK(H2), MEDIAN(H$2:H$394), H2)`

4. Key Findings & Risk Indicators

The overall delinquency rate in this dataset is **17.56%** (69 of 393 customers). We examined the statistical relationships between each feature and the delinquency target, as well as default rates across key customer segments.

Correlation with Delinquent Account

Variable	Correlation	Interpretation
Credit Score	+0.071	Weak positive — counter-intuitive (see warning below)
Annual Income	+0.064	Weak positive
Age	+0.052	Negligible
Loan Balance	+0.051	Negligible
Credit Utilization	+0.048	Negligible
Debt-to-Income Ratio	+0.032	Negligible
Account Tenure	−0.041	Negligible
Missed Payments (12 mo.)	−0.053	Weak negative — counter-intuitive (see warning below)

Delinquency Rates by Customer Segment

Segment	Category	Delinquency Rate
Employment Status	Unemployed	20.83%
	Employed	16.76%
Location	Los Angeles (highest)	23.17%
	New York (lowest)	14.28%
Card Type	Business (highest)	23.75%
	Platinum (lowest)	14.28%

High-Risk Indicators Summary

1. **Unemployment:** Unemployed customers default at a rate 4 percentage points higher than employed customers. Loss of steady income is one of the most reliable early indicators of repayment difficulty.
2. **Los Angeles residency:** Customers in Los Angeles show a 23.17% delinquency rate, nearly 9 points above New York. Higher cost of living in the LA market may be squeezing disposable income and pushing more customers into arrears.
3. **Business card accounts:** Business cardholders default at 23.75%, roughly 10 points higher than Platinum cardholders. This may reflect the higher financial volatility of small business owners compared to established high-credit-score consumers.

⚠ Data Integrity Warning — Counter-Intuitive Correlations

The statistical relationships in this dataset contradict standard credit risk expectations. Delinquent accounts have a *higher* average credit score (598.3) than non-delinquent accounts (567.2), and they show *fewer* recent missed payments (3.99 vs. 4.08 in the last 6 months). In a well-labeled credit portfolio, both of these relationships should be strongly reversed.

Recommendation: Before proceeding with model training, the analytics team should audit the source system that generates the `Delinquent_Account` flag. It is possible that the labels were incorrectly mapped during data extraction, or that this sample was generated with a randomized target variable. Building a model on mislabeled data would produce unreliable risk scores.

5. AI & GenAI Usage

Generative AI tools were used at several stages of this analysis to accelerate the workflow. Below is a summary of the key prompts used and the insights they produced.

Prompt Used	Insight Obtained
"Summarize key patterns, outliers, and missing values in this dataset. Highlight any fields that might present problems for modeling delinquency."	The tool identified the credit utilization values exceeding 100%, flagged the inconsistent employment status labels, and highlighted that three columns contain missing data. It also noted that the `Credit_Card_Type` values overlap with demographic categories.
"Suggest an imputation strategy for missing income values based on industry best practices."	The tool recommended group-based median imputation using employment status as the grouping variable, rather than a single global median. This preserves the income variance across different employment categories.
"Identify the top 3 variables most likely to predict delinquency based on this dataset. Provide brief reasoning."	The tool identified Credit Score, the 6-month payment history (Month_1 through Month_6), and Employment Status as the most relevant predictors. It also flagged the counter-intuitive correlation directions as a potential data quality issue.

6. Conclusion & Next Steps

The Geldium dataset offers a workable starting point for delinquency modeling, but it requires cleaning and validation before it can support reliable predictions. Our analysis uncovered missing values in three critical fields, inconsistent text labels in the employment status column, and outlier credit utilization rates above 100%.

Most importantly, the relationship between the target variable and key credit risk features is statistically unusual. This raises a concern about the accuracy of the delinquency labels themselves, which must be resolved before any model is trained.

Recommended Next Steps

1. **Verify the delinquency labels.** Trace the `Delinquent_Account` flag back to its source system and confirm that the binary values correctly map to actual customer default events.
2. **Standardize categorical fields.** Map `EMP`, `employed`, and `Employed` to a single label. Do the same for `retired` and any other duplicated categories.
3. **Execute imputation.** Apply group-based median imputation for Income and global median imputation for Credit Score and Loan Balance, using the formulas and methods described in Section 3.
4. **Cap credit utilization.** Set an upper bound of 1.0 on the `Credit_Utilization` column to prevent extreme outliers from distorting regression-based models.
5. **Engineer new features.** Create a composite variable counting the total number of “Late” and “Missed” payments across Month_1 to Month_6, as this captures recent payment behavior in a single metric.
